

1

```

src > main.py > ...
29 def main():
    lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    $ python src/main.py
    === MarketPulse ===

    Instrument
    AAPL - Apple Inc.
    Last price: 266.20 USD

    Benchmark
    SP500 - S&P 500
    Last level: 6742.00

    Period: 1 month
    Interval: Daily

    Observations
    AAPL: 21
    SP500: 21
    <class 'dict'>
    dict_keys(['instrument', 'benchmark'])
    AAPL
    Apple Inc.
    USD
    SP500
    S&P 500

    lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    $ head data/sample/prices.csv
    date,ticker,open,high,low,close,volume
    2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
    2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
    2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
    2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
    2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
    2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
    2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
    2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
    2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000

    lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    $ head data/sample/prices.csv
  
```

2

```

src > main.py > ...
29 def main():
    lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    $ head data/sample/prices.csv

    lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    $ head data/sample/prices.csv
    date,ticker,open,high,low,close,volume
    2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
    2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
    2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
    2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
    2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
    2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
    2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
    2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
    2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000

    lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    $ python src/main.py
    === MarketPulse ===

    Instrument
    AAPL - Apple Inc.
    Last price: 266.20 USD

    Benchmark
    SP500 - S&P 500
    Last level: 6742.00

    Period: 1 month
    Interval: Daily

    Observations
    AAPL: 21
    SP500: 21
    <class 'dict'>
    dict_keys(['instrument', 'benchmark'])
    AAPL
    Apple Inc.
    USD
    SP500
    S&P 500
  
```

```

def main():
    # Fetch instrument details
    instrument = get_instrument('AAPL')
    print(f'Instrument: {instrument["name"]} - {instrument["symbol"]}')
    print(f'Last price: {instrument["last_price"]} USD')

    # Fetch benchmark details
    benchmark = get_benchmark('SP500')
    print(f'Benchmark: {benchmark["name"]} - {benchmark["symbol"]}')
    print(f>Last level: {benchmark["last_level"]} USD')

    # Fetch observations
    observations = get_observations('AAPL')
    for observation in observations:
        print(f'Date: {observation["date"]} | Ticker: {observation["ticker"]} | Open: {observation["open"]} | High: {observation["high"]} | Low: {observation["low"]} | Close: {observation["close"]} | Volume: {observation["volume"]}')

if __name__ == '__main__':
    main()

```

The screenshot shows a VS Code editor with a file explorer on the left, a terminal at the bottom, and a code editor in the center. The file explorer shows a project structure with files like README.md, CONTRIBUTING.md, and TEAM.md. The code editor shows the main.py file with the following content:

```

src > main.py > ...
79 def main():

Problems Output Debug Console Terminal Ports
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.28 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

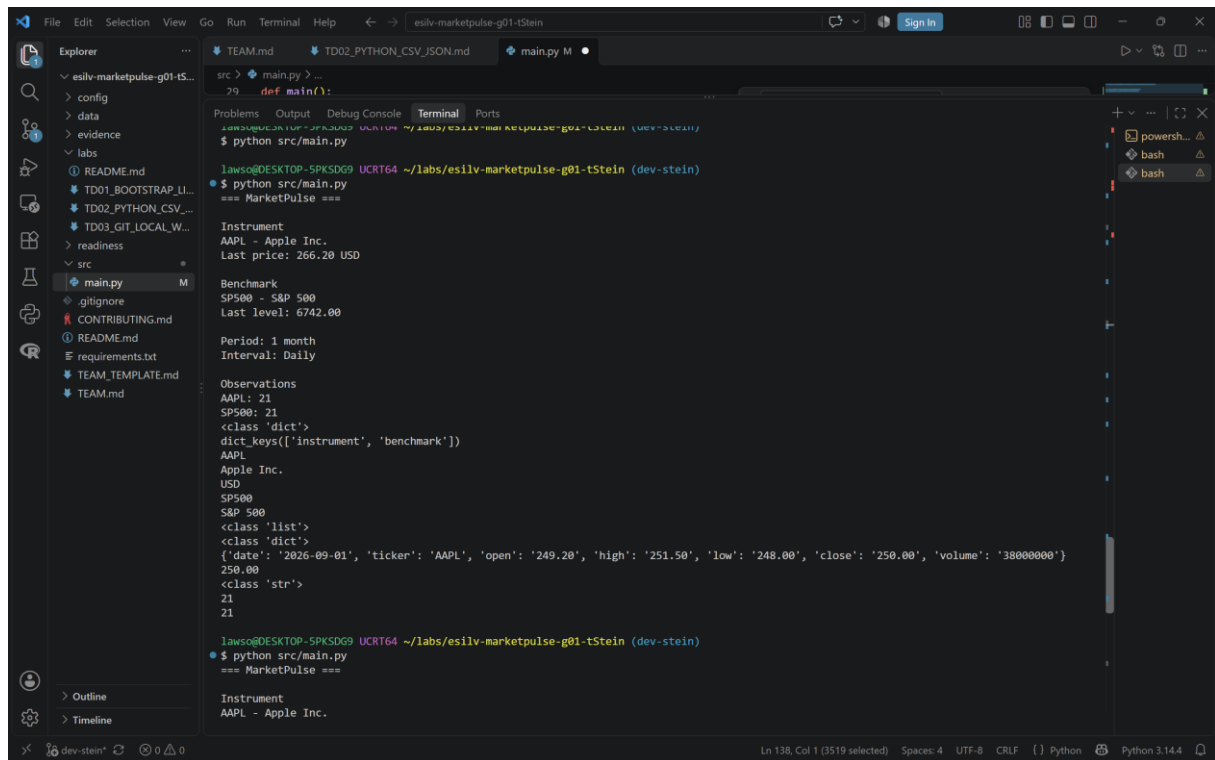
Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
AAPL
Apple Inc.
USD
SP500
S&P 500
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.28', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '380000000'}
250.00
<class 'str'>

lawso@DESKTOP-5PKSDG9 UCR164 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
$ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.28 USD

```



```
src > main.py > ...
29 def main():
    """
    """
    # Get the instrument and benchmark
    instrument = AAPL
    benchmark = SP500

    # Get the observations
    observations = get_observations(instrument, benchmark)

    # Print the results
    print(f"Instrument: {instrument}")
    print(f"Benchmark: {benchmark}")
    print(f"Observations: {observations}")

    # Return the results
    return {"instrument": instrument, "benchmark": benchmark, "observations": observations}

if __name__ == "__main__":
    main()

lawso@DESKTOP-SPKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
$ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
AAPL
Apple Inc.
USD
SP500
S&P 500
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
250.00
<class 'str'>
21
21

lawso@DESKTOP-SPKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
$ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
```

```
src > main.py > ...
29 def main():
    lawso@DESKTOP-5PKSDG9 UCR164 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    $ python src/main.py
    lawso@DESKTOP-5PKSDG9 UCR164 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    $ python src/main.py
    === MarketPulse ===

    Instrument
    AAPL - Apple Inc.
    Last price: 266.20 USD

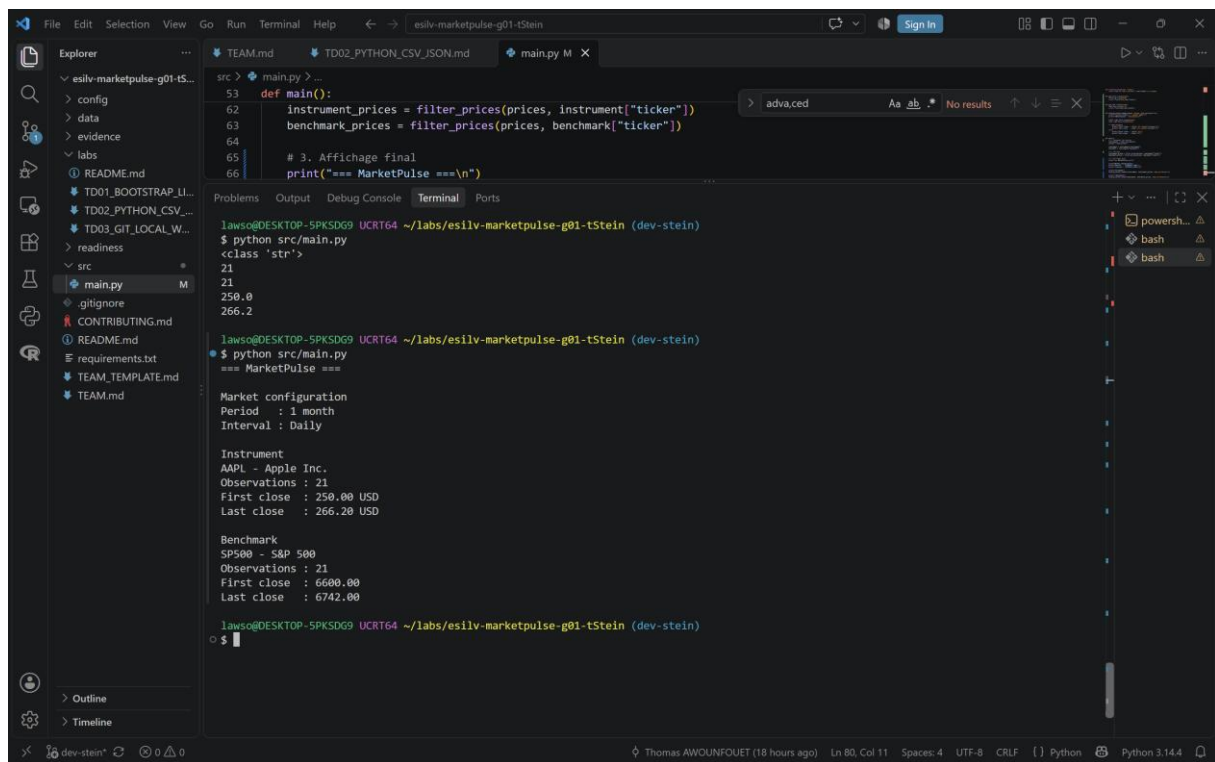
    Benchmark
    SP500 - S&P 500
    Last level: 6742.00

    Period: 1 month
    Interval: Daily

    Observations
    AAPL: 21
    SP500: 21
    <class 'dict'>
    dict_keys(['instrument', 'benchmark'])
    AAPL
    Apple Inc.
    USD
    SP500
    S&P 500
    <class 'list'>
    <class 'dict'>
    {'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
    250.00
    <class 'str'>
    21
    21
    250.0
    266.2

    lawso@DESKTOP-5PKSDG9 UCR164 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)
    o $
```

6-9



The screenshot shows a Visual Studio Code editor window with the following components:

- Explorer:** A file tree on the left showing the project structure for 'esilv-marketpulse-g01-tStein'. It includes folders like 'config', 'data', 'evidence', 'labs', and files like 'README.md', 'TD01_BOOTSTRAP_LI...', 'TD02_PYTHON_CSV...', 'TD03_GIT_LOCAL_W...', 'readiness', 'src', 'main.py', '.gitignore', 'CONTRIBUTING.md', 'requirements.txt', 'TEAM_TEMPLATE.md', and 'TEAM.md'.
- Editor:** The main workspace shows the 'main.py' file. The code defines a 'main()' function that filters prices for an instrument and a benchmark, then prints a 'MarketPulse' summary. The code is as follows:

```
53 def main():  
62     instrument_prices = filter_prices(prices, instrument["ticker"])  
63     benchmark_prices = filter_prices(prices, benchmark["ticker"])  
64  
65     # 3. Affichage final  
66     print("=== MarketPulse ===\n")
```
- Terminal:** The bottom panel shows the execution of the script. The command 'python src/main.py' is run, and the output is:

```
lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)  
$ python src/main.py  
<class 'str'>  
21  
21  
250.0  
266.2  
  
lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)  
$ python src/main.py  
=== MarketPulse ===  
  
Market configuration  
Period : 1 month  
Interval : Daily  
  
Instrument  
AAPL - Apple Inc.  
Observations : 21  
First close : 250.00 USD  
Last close : 266.20 USD  
  
Benchmark  
SP500 - S&P 500  
Observations : 21  
First close : 6600.00  
Last close : 6742.00  
  
lawso@DESKTOP-5PKSDG9 UCR764 ~/labs/esilv-marketpulse-g01-tStein (dev-stein)  
$
```